

Competition Policy for Informative Market Access

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Abstract

Competition policy changes both productive opportunities and the evidence available for consequential decisions. This paper develops a remedy framework that begins with a defined service and decision boundary rather than an assumed market-power category. Eligibility, routing, entry costs, and reinforcement generate contestability; a comparison budget diagnoses exposure but does not itself rank welfare. Policies are evaluated through operating surplus, implementation costs, and discounted decision losses across a documented set of plausible models. A common decision rule is essential: each model can assign substantial value to a signal while uncertainty about its interpretation eliminates every implementable informational gain. We characterize robust remedy certificates through a finite linear program, derive a decision-specific accuracy threshold, and show how simultaneous model coverage supports selection after calibration. Trusted metadata and trial outcomes can be statistical complements. A transparent hypothetical procurement illustration compares audits, portability, interoperability, pilots, and replications by certified benefit and cost. Existing European antitrust and digital regulation retain their statutory conditions; a separate prospective evaluation mandate would require explicit authorization and safeguards. The contribution is an auditable bridge from comparison opportunities to lawful, costed decisions under partial knowledge.

Keywords: competition policy, informative access, robust decisions, comparison budget, remedy design, partial identification.

JEL codes: D61, D81, D83, L41, L51, C18.

1. Introduction

Access rules determine more than which firms sell. They determine which alternatives are encountered, which users can switch without losing relevant histories, and which interventions leave interpretable evidence. A procurement platform may display an established supplier repeatedly while recording almost no outcomes for comparable entrants. Another platform may concentrate transactions in one service but run carefully calibrated comparisons among eligible suppliers. These arrangements can have similar aggregate shares and very different implications for later decisions. The relevant economic question is which comparisons improve a specified decision enough to justify their full costs.

Existing competition analysis already considers entry, innovation, data advantages, and dynamic efficiency. The missing step addressed here is more precise: an authority must choose an implementable remedy and a subsequent decision rule when it does not know the correct observation model. A finding that additional data would help an analyst who knew that model is insufficient. The rule actually used must interpret the records correctly across the models that remain plausible. Missing calibration, incompatible definitions, or uncertain transportability can prevent a nominal information improvement from delivering an attainable welfare gain.

Shadow Futures supplies one motivation for investigating this problem (Erlic, 2026). In a specified adaptive allocation experiment, residual contestability is the conditional probability of allocation outside the current leader. Its cumulative sum is a comparison budget. With bounded input differences, that budget bounds conditional Fisher information and finite-horizon statistical separation. Under a common predictable design, local equivalence, and comparison-dominated separation, an almost-surely finite budget prevents universally consistent recovery of a

nonconstant contribution functional from one complete history. These statistical restrictions do not price missing comparisons or determine policy.

This article builds the decision layer. It defines a service boundary and a payoff-relevant decision, represents access policies as costed experiments, and retains uncertainty about their signal distributions and economic effects. Its central object is a robust certificate: one rule whose expected incremental welfare exceeds a declared threshold in every retained model. A finite linear program computes that certificate. An accuracy inequality makes the same logic measurable in a binary application. Neither requires concentration or a market-power designation as an economic primitive.

Three results give the framework practical content. First, a common-rule certificate and its dual identify the adverse model mixture against which a proposed remedy must work. Second, a polarity example shows why substantial model-specific information value can disappear without trusted interpretation; calibration metadata and observations may be complements rather than substitutes. Third, a simultaneous model-coverage argument explains when selecting a remedy after calibration preserves a stated confidence guarantee for its prospective expected welfare. A finite-cost bound also explains why an efficient pilot need not pursue universal identification forever.

The intended use is prospective remedy design, procurement evaluation, or economic evidence within an existing legal route. The framework does not establish an information externality merely by showing an inference limit. A public intervention additionally requires a defensible objective, incomplete internalization or another justification, and costs below the benefits it can actually attain. European law supplies separate requirements for antitrust liability, gatekeeper

designation, and systemic-risk research access. These are treated as constraints on the available policy set.

The contribution is therefore neither a new general Blackwell theorem nor a claim that large firms destroy knowledge. It is a decision-specific account of informative access under partial knowledge, with certificates that can fail, an explicit remedy comparison, and a proposed institutional procedure for testing the assumptions on which intervention depends. Sections 2–4 establish its economic and statistical foundations; Sections 5–8 derive and illustrate the certificate; Sections 9–11 address implementation, legal fit, and limitations. The appendices give proofs and reproducible calculations.

2. Related literature and the contribution

Classical contestability concerns the discipline exerted by potential entry and the implications of entry and exit conditions for industry structure (Baumol, 1982). The present object is different. A threat of entry may constrain prices without producing observations that discriminate among technologies; actual trials may be informative without disciplining every price. Informative access is consequently a decision-specific attribute of a service arrangement, not a synonym for perfect contestability. Scale and compatibility benefits remain part of welfare rather than obstacles to be assumed away (Katz and Shapiro, 1985).

Information economics supplies the foundational comparison. Blackwell (1953) relates experiments through state-independent garbling and their value in fixed decision problems. Arrow (1962) connects the production and appropriation of knowledge to economic welfare. Markets for information concern its design, sale, filtering, and possible adverse uses (Bergemann and Bonatti, 2019). Our certificates use these insights while accounting separately for operating effects and intervention costs. More informative records have nonnegative gross value under the

usual decision assumptions; that result does not establish a positive net return to an access obligation.

Robust evaluation also has established foundations. Gilboa and Schmeidler (1989) characterize maxmin expected utility with multiple priors. Wang (2024) develops informativeness orders for ambiguous experiments. This article does not claim to generalize those orderings. It specializes a common-rule robust decision problem to a finite remedy menu, makes its welfare threshold explicit, and connects the certificate to evidence collection and legal implementation. The polarity example isolates a practical failure of model-specific valuation, rather than presenting ambiguity about experiments as a new discovery.

Decision theory and experimental design distinguish useful policy learning from estimation for its own sake. Manski (2004) studies treatment rules and finite-sample regret; Kasy and Sautmann (2021) study adaptive assignment for policy choice. We similarly value a comparison by the decision it can change. Signal selection can itself be strategic, as Bayesian persuasion makes clear (Kamenica and Gentzkow, 2011). Registration and independent calibration are therefore economic design features, not administrative decorations. In strategic environments, information affects incentives as well as inference (Bergemann and Morris, 2016); those effects must enter the candidate models.

Data and competition are likewise an existing literature. Farboodi et al. (2019) model data accumulation and firm dynamics; Jones and Tonetti (2020) examine nonrival data, privacy, and inefficient hoarding. Crémer, de Montjoye, and Schweitzer (2019) analyze digital competition, platforms, and access to data. This article's narrower addition is an implementability requirement for information-based remedies: disclose the action, common interpretation rule, model set, and

full cost threshold through which an access change is claimed to improve welfare. Counting records, entrants, or connections does not meet that requirement.

3. The service boundary and endogenous comparison

Specify the service task before assessing informative access. The boundary identifies affected users, eligible alternatives, geography, time horizon, and the subsequent decision. For a procurement application it might encompass the routing of qualified suppliers to municipal service trials over a year, followed by selection of a contract design. Register the eligibility conditions, observables, and feasible downstream actions. This is an observation and intervention boundary. A competition-law relevant market requires its own substitution analysis and cannot be inferred from this description.

Policy r affects entry costs, verification, transferable histories, interoperability, and routing. In one allocation model, the eligible set $I_t(r)$ and verified input profile $x_{i,t}$ determine opportunities together with accumulated position $s_{i,t}$. For payoff state ω and model m , a structural coefficient $\beta(\omega)$ enters conditional odds while the model specifies feedback and other uncertain features:

$$p_{i,t}(\omega, m; r) = \frac{\exp(x_{i,t}'\beta(\omega) + s_{i,t}(r, m))}{\sum_{j \in I_t(r)} \exp(x_{j,t}'\beta(\omega) + s_{j,t}(r, m))},$$

$$s_{t+1} = F_{r,m}(s_t, J_{t+1}, x_t),$$

$$B_T(\omega, m; r) = \sum_{t < T} [1 - \max_i p_{i,t}(\omega, m; r)].$$

Contestability is generated by these primitives and by participants' responses. The policy may change both the alternatives willing to enter and the chance that an eligible alternative is tried.

Concentration is an outcome to measure, not a sufficient statistic for the available comparisons.

An increasing budget does not guarantee useful input variation, and a decreasing budget does not

establish a Blackwell ordering of complete experiments. The budget is a diagnostic within a specified allocation model.

Counterexample 1 makes the distinction exact. A hypothetical provider serves 95 percent of users but independently allocates trials between two eligible suppliers. Let $x_1 - x_2 = 1$, both state indices be zero, and β belong to $\{-b, b\}$, with $b > 0$. Then the recipient probability for supplier 1 is logistic in β and the budget grows linearly:

$$\Pr_{\beta}(J_{t+1} = 1) = \frac{1}{1 + \exp(-\beta)},$$

$$B_T = \frac{T}{1 + \exp(b)}.$$

The two recipient distributions differ, and independent sample frequencies consistently identify the sign of β . A large provider share is compatible with indefinitely informative supplier comparisons. Conversely, several providers can share one ranking system, observation database, and common shocks. Organizational multiplicity then need not produce independent replication. Neither example resolves a legal dominance inquiry or establishes efficient industry structure.

An experiment must include all observations actually used for inference. Inputs, eligibility events, allocation recipients, outcomes, versions, and legitimately observed propensities may each carry information. Parameter-dependent records cannot be ignored to obtain an impossibility conclusion. Nor can an unobserved propensity be treated as known because it would make a preferred estimator work. Auditors must explain how probabilities are measured or bounded independently of the outcome pattern they seek to explain.

Intervention stability is equally necessary. Portability may change who enters; interoperability may change the users reached; a pilot may change effort or learning. The same β need not describe all these environments. Models may retain a common parameter only with justified invariance or transportability restrictions. Where those restrictions are weak, the remedy problem

retains several plausible mappings from trial observations to downstream performance rather than concealing the uncertainty.

4. Welfare under partial knowledge

Let Ω be a finite set of payoff-relevant states, A a common finite action set, and R a finite menu of feasible policies, including baseline 0. The state may describe which supplier technology performs better for a registered task. For each r , the finite signal space Z_r records the evidence made available by the policy. Finite spaces permit exact computation; Appendix A explains how a bounded approximation error can extend a certificate to a larger model class.

A model m specifies a prior π_m on Ω , an observation kernel $K_{r,m}(z|\omega)$, a bounded loss $\ell_m(a,\omega)$, expected operating surplus $S_{r,m}$, and additional implementation costs $C_{r,m}$. The set M contains the joint models retained after evaluating the available evidence. It can represent uncertainty about demand, calibration, entry response, investment effects, and welfare stakes. Restrictions connecting these objects matter: independently choosing the worst component of each can construct an economic environment that no retained model permits.

A decision rule d assigns probabilities $d(a|z)$ to actions after a signal. It is one implementable mapping, shared across models. If observed metadata can reveal the environment, include it in z ; the rule may use it. It may not condition on an unobserved model label. This distinction separates a real evaluation procedure from an analyst's oracle who silently chooses the best interpretation for whichever model happens to be true.

Operating surplus records productive and allocative effects over the stated evaluation horizon. Additional costs include implementation, auditing, compliance, privacy, security, duplicated resources, and changes in investment incentives. The downstream loss concerns decisions after review at date L . Each effect belongs in one part of the account. Monetary transfers cannot also

be counted as destroyed resources, and future productive benefits represented in the decision loss cannot be added again as operating gains. With discount factor $0 < \delta \leq 1$, expected welfare is:

$$J_m(r, d) = S_{r,m} - C_{r,m} - \delta^L \sum_{\omega, z, a} \pi_m(\omega) K_{r,m}(z|\omega) d(a|z) \ell_m(a, \omega).$$

Define a common existing rule d_0 and a lawful prospective baseline. The incremental welfare $\Delta_m(r, d)$ is $J_m(r, d) - J_m(0, d_0)$. The robust value q_r is the best lower guarantee attainable with one rule. A positive q_r is a model-relative expected-welfare statement; it is not a guarantee that every realized trial or procurement outcome will be beneficial.

$$\Delta_m(r, d) = J_m(r, d) - J_m(0, d_0),$$

$$q_r = \max_{d \in D_r} \min_{m \in M} \Delta_m(r, d).$$

A robust criterion is a declared policy judgment, not a claim that public authorities must be infinitely ambiguity averse. Expected welfare under an explicit model weighting or minimax regret can be reported alongside q_r . The virtue of a lower certificate is narrower: its promise does not depend on selecting a favorite model after seeing which remedy looks best. A larger model set generally weakens that promise. Missing evidence can therefore change the appropriate intervention without implying that existing mandatory duties disappear.

The loss function identifies whose interests count and the consequences of a mistaken decision. The relevant stakes may concern reliability, accessibility, procurement cost, or a regulated safety outcome; they need not be retrospective judgments about merit. If an experiment improves decisions for dispersed beneficiaries while its sponsor cannot appropriate those benefits, an information externality may exist. Incomplete contracting, spillovers, and private experimentation must be investigated separately. A finite comparison budget establishes none of them.

Finally, the procedure commits ex ante to a rule or to a registered update mechanism.

Reoptimizing different maxmin problems after each signal can introduce a different ambiguity

model and different incentives. Dynamic consistency requires additional structure. The finite problem here concerns a single review decision, with any staged response represented in the registered action or policy menu. This keeps the formal certificate aligned with a procedure that could actually be administered.

5. Common-rule certificates and calibration

Proposition 1 gives an exact computational test. For a fixed policy r and finite M , define

$h_{r,m}(z,a) = \sum_{\omega} \pi_m(\omega) K_{r,m}(z|\omega) \ell_m(a,\omega)$, and $b_{r,m} = S_{r,m} - C_{r,m} - J_m(0, d_0)$. Then q_r is the optimum of the following linear program:

$$\begin{aligned} & \max_{v,d} v \\ & v \leq b_{r,m} - \delta^L \sum_{z,a} h_{r,m}(z,a) d(a|z) \quad (m \in M), \\ & d(a|z) \geq 0, \quad \sum_a d(a|z) = 1 \quad (z \in Z_r). \end{aligned}$$

A feasible solution with $v \geq \tau$ is a certificate that one common rule achieves the target τ in every retained model. Maximizing separately under each model supplies no such certificate.

Randomized rules are allowed because randomization can improve a worst-case guarantee.

Where law or administration prohibits random decisions, restrict D_r to deterministic rules and enumerate them; the unrestricted linear program then gives an upper bound rather than an implementable optimum.

The dual has an informative interpretation. It chooses weights w on the retained models to construct the adverse mixture against which even the best common rule performs least well:

$$q_r = \min_{w \in \Delta(M)} \left[\sum_m w_m b_{r,m} - \delta^L \sum_z \min_{a \in A} \sum_m w_m h_{r,m}(z,a) \right].$$

These weights are a mathematical witness, not estimates of model probabilities. They identify an economically relevant failure mode that can guide additional measurement. If the adverse mixture concentrates on models differing only in calibration, a calibration audit may be more

useful than additional observations of the same uncertain signal. If it concentrates on investment-cost models, better cost evidence is the relevant next step. The proof uses ordinary finite minimax duality; the policy contribution lies in the certificate's required economic inputs. Proposition 2 isolates an implementability gap. Let ω be uniformly distributed on $\{0,1\}$, let wrong action cost one, and let baseline evidence be uninformative. Under model $+$, signal z equals ω with probability $q > 1/2$. Under model $-$, it equals $1-\omega$ with the same probability q . An analyst told the model achieves risk $1-q$ in either case. Any rule using z without trusted orientation has worst-case risk at least $1/2$, attained by ignoring it. Thus each model reports a strict gross benefit $q-1/2$, but no common rule attains a positive gross benefit.

The proof is immediate but consequential: the two risks sum to one for every randomized rule, so their maximum is at least one-half. A positive cost makes the uncalibrated intervention strictly worse than baseline. More records cannot resolve a signal-label ambiguity that leaves the observable signal distribution unchanged. This is not a claim that information intrinsically harms a decision maker who can preserve and ignore it; it is a claim that model-specific usefulness is weaker than implementable usefulness.

If a trusted metadata field reveals whether the model is $+$ or $-$, a single rule can follow z in the first case and reverse it in the second. Metadata alone leaves risk one-half; the raw signal alone also leaves robust risk one-half; together they give risk $1-q$. They are statistical complements. The implication is to evaluate bundles that combine interpretable definitions, provenance, and observations, rather than valuing each access item by its standalone information content.

A sharper version concerns replacement. Suppose existing evidence is a known, correctly oriented signal of accuracy 0.75, while a replacement is the orientation-ambiguous signal with $q=0.9$. Within either known model the replacement can be garbled into the old experiment.

Nevertheless its common-rule worst-case risk is 0.5, compared with 0.25 for the existing rule.

Model-specific garblings do not justify replacing a trusted interface. If both experiments remain available, the old rule can be retained; preservation of its interpretation is the relevant design safeguard.

These examples sit within the established literature on ambiguous experiments, rather than competing with its general characterizations (Wang, 2024). Their substantive implication is a measurable standard for policy claims: specify what calibrates the signal, whether that calibration survives the remedy, and whether every model used in the appraisal permits the same registered decision rule to exploit it.

6. From evidence to a decision-specific threshold

A certificate needs evidence about complete models, not only a transaction count. Begin with the decision and a lawful comparator. State the observation registry, the intervention contrast, and the loss from incorrect choice. Register candidate models before choosing a preferred remedy. Document which data constrain each model component and which restrictions remain structural assumptions. Claims about independence, transportability, or an unobserved allocation rule should be marked as such.

A useful simple case has binary states and actions. Baseline rule d_0 always chooses action 0. Let $\pi = \Pr(\omega=1)$ lie in $[\pi_L, \pi_U]$. A correctly oriented symmetric signal under remedy r has accuracy $q \geq q_L$. Wrong choice costs $\lambda > 0$, and the rule follows the signal. Baseline expected loss is $\lambda\pi$, while the new rule's loss is $\lambda(1-q)$, regardless of π . Suppose $\Delta S \geq s_L$ and $\Delta C \leq c_U$. Every model satisfying these restrictions then gives:

$$\Delta_m(r, d_{\text{follow}}) \geq s_L - c_U + \delta^L \lambda (\pi_L + q_L - 1).$$

Proposition 3. Under these assumptions, the displayed quantity is a sufficient lower certificate. If it is at least τ , following the signal is a feasible common rule meeting the target. If the uncertainty set includes the joint corner $\pi=\pi_L$, $q=q_L$, $\Delta S=s_L$, and $\Delta C=c_U$, the bound is exact for that rule. An optimized robust rule can do better, so failure of this sufficient bound is not proof that every rule or remedy fails.

Solving the inequality gives a break-even accuracy rather than an unexplained requirement for more competition:

$$q_L \geq 1 - \pi_L + \frac{\tau - s_L + c_U}{\delta^L \lambda}.$$

If the right side exceeds one, this rule cannot meet the target even with perfect information under the stated costs and stakes. If it lies below the current guaranteed accuracy, a cheap audit or an operating-efficiency change may suffice. The threshold separates the evaluation objective from an arbitrary exposure floor. Asymmetric error losses or different baseline rules alter the expression and should be modeled rather than forced into the symmetric case.

Accuracy restrictions require a defensible calibration design. Historical tasks with independently established outcomes may constrain a signal's error rates. They must represent the downstream decision or support a stated transportability bound. Training and calibration on the same selected records can overstate accuracy. Unobserved outcomes for rejected alternatives cannot simply be assigned favorable labels. Selection corrections, random assignment, or partial-identification bounds must correspond to the particular missingness process.

For allocation evidence, the registry should include eligibility, relevant input differences, routing propensities when independently known, treatment versions, and outcome definitions. A comparison probability near zero diagnoses a missing contrast; it does not reveal how valuable that contrast would have been. Where a propensity cannot be observed, an identified interval can

be propagated through the candidate kernels. Additional parameter-dependent information belongs in those kernels, including any audit measurement used to estimate the propensity itself. Proposition 4 addresses policy selection after calibration. Let D be calibration evidence from tasks independent of the future payoff state and future experiment conditional on the true model. Suppose $M(D)$ is a simultaneous confidence set containing that model with probability at least $1-\alpha$. If the procedure selects $r(D), d(D)$ satisfying $\Delta_m(r(D), d(D)) \geq \tau$ for every $m \in M(D)$, then with probability at least $1-\alpha$ its conditional expected future welfare gain is at least τ . Appendix A supplies the proof and its limits.

The claim concerns prospective expectation, not guaranteed realized benefits. Coverage must survive model selection, multiple candidate remedies, and any sequential stopping used in calibration. Separate pointwise intervals assembled after favorable outcomes do not establish simultaneous coverage. Structural misspecification and changes in the future environment can defeat the guarantee even when sampling uncertainty is controlled. The strongest defensible statement may therefore be conditional on a restricted environment and an explicit transportability allowance.

7. Costed mechanisms and least-cost choice

Data access and auditing reveal existing records and can test eligibility, routing, or calibration assumptions. Their incremental value depends on the omitted contrast. Secure access may be the least costly route where the relevant observations already exist. It cannot recreate an alternative allocation history. An audit that only confirms the incumbent's selected successes may lower uncertainty about its own reliability while leaving the rival comparison unidentified. Scope the requested fields to the decision and include confidentiality, privacy, and security costs.

Portability changes usable histories and switching conditions. It may preserve verification, reduce entrants' credential costs, or let users compare services without forfeiting accumulated information. Identical copies of a record are not independent replications. Semantically incompatible exports may also reduce the value of previously trusted evidence, as the replacement example shows. Provenance and common definitions can be more valuable than additional file volume. User selection into switching and changes in effort belong in the candidate models.

Interoperability changes feasible interactions and access to complementary users or features. Compatibility can create real network benefits (Katz and Shapiro, 1985), but connections do not automatically produce a new experiment. Shared routing, common control, correlated shocks, and strategically altered participation may leave the crucial contrast unchanged. Interface specification, security, capacity, and innovation responses determine its costed welfare effect. A usable experiment may require a narrow interface rather than general reciprocal access.

Randomized pilots deliberately generate the missing contrast. Their design should target future policy choice, not a statistically significant effect on any convenient outcome (Kasy and Sautmann, 2021). Assignment probabilities, eligibility, endpoints, and stopping should be registered. Spillovers may require cluster designs or a model of interference. Opportunity costs to trial participants and production displaced during the pilot are operating effects; audit and setup expenses are additional costs. They must not be charged twice.

Independent replications provide fresh shocks under a common informative experiment. Separate legal entities do not establish statistical independence. Verify distinct assignment, relevant conditional independence, and a common parameter or justified transportability. Replication can be wasteful where it merely duplicates the same correlated history. It can be useful where a fresh

informative environment resolves ambiguity that no extension of one reinforced path can remove. Real scale economies and the cost of maintaining duplicated infrastructure remain in the welfare account.

Define burden k_r before comparing remedies; for example, take the worst retained implementation cost. For a justified positive target τ , select the least burdensome lawful policy whose optimized common-rule certificate reaches the target:

$$r^* \in \operatorname{argmin}_{r \in R_{\text{legal}}} k_r \text{ subject to } q_r \geq \tau, \quad k_r = \max_{m \in M} C_{r,m}.$$

The criterion differs from maximizing robust welfare, and the objective must be stated. A public authority may seek the least costly way to meet a statutory performance objective rather than the largest available gain. Report both where useful. Bundles belong in R because calibration and observations can complement one another. A tax, subsidy, or access quota cannot be treated as equivalent without modeling the behavior it induces; instrument uncertainty is an established problem (Weitzman, 1974).

Proposition 5 sharpens pilot sizing in a symmetric binary experiment. Suppose trial accuracy p is known only to satisfy $p \geq p_L$ with $1/2 < p_L < 1$, n is odd, operating gains are $H_0 - \eta n$, and implementation costs are $F + \kappa n$, with $\eta \geq 0$, $\kappa > 0$, and $F \geq 0$. Assume the uncertainty set permits the lower-accuracy and lower-prior corner. The follow-majority rule has lower certificate Γ_n below. Its gains from adding two trials strictly decrease with n . Among pilots committing to this rule, a certificate-maximizing positive pilot stops when the next two-trial gain falls below their combined marginal cost. Setup cost affects participation, while decision stakes affect efficient intensity.

$$\Gamma_n = H_0 - F - (\eta + \kappa)n + \delta^L \lambda [\pi_L + q_n(p_L) - 1].$$

The binomial calculation and stopping inequality appear in Appendix B. Conditional on selecting a positive pilot, higher decision stakes or lower per-trial costs weakly increase the preferred trial

count in this restricted model. More accurate individual trials need not do so: their additional-sample gain can be small because one trial already resolves most of the uncertainty. This distinction prevents a blanket recommendation to maximize either exposure or sample size. Measurement intensity is also a choice. If a pilot with n trials costs at least κn , $\kappa > 0$, while all possible operating gains are bounded above by H and its discounted decision improvement by $\delta^L \lambda$, no pilot with $\kappa n > H + \delta^L \lambda$ can beat the costless baseline. Proposition 6 therefore bounds the efficient trial count whenever this menu and accounting apply. A demand for universal identification is stronger than the welfare objective. Finite experimentation can be efficient even when a statistical theorem denies learning every structural parameter from one path.

8. A transparent procurement illustration

Consider a hypothetical public purchaser choosing between two qualified service designs for the following year. Action 0 is the existing contract design; action 1 is an alternative whose relative suitability is uncertain. A wrong choice has loss $\lambda = \text{€}1$ million over that review horizon. The authority retains $\pi \in [0.45, 0.55]$ for the probability that action 1 is better. Discounting gives $\delta^L = 0.9$. These are stipulated decision units and uncertainty restrictions, not estimates from a real purchaser, supplier, or platform.

Five candidate policies have correctly oriented signal-accuracy lower bounds, operating-gain lower bounds, and implementation-cost upper bounds. The bounds are deliberately expressed in money and error probabilities that an actual appraisal would have to justify. The illustration applies Proposition 3 with the common follow-signal rule. It reports a sufficient attainable guarantee, rather than the optimized q_r of every possible rule. Costs are assumed to include the nonoperating burdens described earlier.

Table 1 Hypothetical common-rule remedy certificates

Policy	Accuracy lower bound	Operating gain lower bound €m	Implementation cost upper bound €m	Welfare gain lower certificate €m
Scoped audit	0.6000	0.05	0.06	0.0350
Usable portability	0.6500	0.03	0.10	0.0200
Specified interoperability	0.7500	0.04	0.18	0.0400
Registered 15-trial pilot	0.7869	-0.03	0.08	0.1032
Independent 31-trial replication package	0.8716	0.00	0.22	0.0695

All inputs are hypothetical. Certificates use $\Delta W \geq s_L - c_U + 0.9(0.45 + q_L - 1)$. The pilot and replication accuracy values come from the exact symmetric Bernoulli experiment in Appendix B; displayed values are rounded.

At a target $\tau = \text{€}30,000$, the scoped audit has the smallest stipulated implementation burden among these certified policies. Portability misses the target; interoperability, the pilot, and replication meet it at greater cost. At $\tau = \text{€}60,000$, the audit and interoperability miss it, while the pilot is less burdensome than the replication package. The optimized linear program could certify additional rules, but these comparisons already demonstrate that the appropriate instrument depends on the target rather than a universal ranking of access policies.

The pilot's accuracy comes from an explicit experiment. Conditional on state 1, each registered independent trial produces a binary signal with success probability 0.6; conditional on state 0, that probability is 0.4. For odd n , the authority chooses state 1 after a majority of successes. Its accuracy is the upper binomial tail at 0.6. Fifteen trials give 0.7868968; thirty-one give 0.8716183. The numbers follow from the stipulated kernel, not an empirical estimate or a claim that every field pilot has this distribution.

Suppose the pilot is only partly transportable. With probability at most ρ , its signal is replaced by an arbitrary signal unrelated to the downstream contract. In that contamination model, its accuracy guarantee falls to at least $(1-\rho)q(n)$. At $\rho=0.15$, the fifteen-trial lower accuracy becomes about 0.6689; its welfare certificate falls from about €103,207 to about –€3,024. This shows why a convincing local contrast can fail to justify a downstream remedy when transportability is weak.

The comparison also exposes measurement priorities. The pilot reaches the €60,000 target only if $q_L \geq 0.7389$ under its stipulated operating and implementation bounds. A calibration exercise that could establish only $q_L = 0.70$ would not support that certificate. By contrast, reducing its cost upper bound from €80,000 to €40,000 lowers the required accuracy by about 0.0444. Better evidence about costs and better evidence about signal validity are alternative ways to resolve the same decision threshold.

An actual design would first record the existing procurement rule and define the task on which relative suitability will be assessed. Eligible suppliers would meet safety and reliability constraints before random assignment. The pilot would register endpoints, version controls, assignment probabilities, interference assumptions, data retention, and the future contract rule. Calibration tasks would be distinct from the final decision tasks. Joint confidence bounds would cover the accuracy, operating effects, and costs used in certification.

Neither the audit nor the pilot establishes general merit rankings or an entitlement to future contracts. The output is a conditional comparison between specified lawful procedures. Its empirical content is falsifiable: a purportedly portable history must remain interpretable, the accuracy restriction must survive validation, the cost bound must cover actual implementation,

and the registered decision must use the promised evidence. The case illustrates an appraisal method whose entries can be challenged rather than an anecdote offered as evidence of harm.

9. Institutional procedure and evidentiary standards

A practical appraisal should distinguish four evidentiary stages. The first establishes the service boundary and missing comparison. The second identifies a decision and a signal model through which a feasible intervention could improve it. The third quantifies operating effects, implementation costs, and attainable decision losses. The fourth identifies the legal authority and conditions for acting. A failure at an earlier stage cannot be repaired by precision at a later one: a detailed cost estimate does not validate an uninterpretable signal.

At the first stage, identify which relevant alternatives are eligible, observed, excluded, or missing. Report measured propensities or justified intervals, input variation, and any channels through which substitute evidence already arrives. Concentration measures can be useful descriptive evidence but do not establish the comparison kernel. A finding that routing eliminates a contrast should specify the parameter or decision it obstructs and the period over which that obstruction matters.

At the second stage, disclose the common rule and calibration restrictions. Explain whether existing evidence is preserved, supplemented, or replaced. The polarity example makes semantics testable: if changing a field definition changes the direction in which a rule responds, trusted metadata are part of the experiment. Require independent verification of important labels and version changes. Where firms select which records are released, model that selection rather than treating disclosure as a neutral sample.

At the third stage, submit the remedy menu, joint model set, welfare weights, and break-even thresholds. Include credible scale benefits and the possibility that existing actors already

internalize useful experimentation. Show how the certificate changes when uncertain assumptions are relaxed. A positive result under one narrow specification is evidence about that specification; a robust result states what survives the retained alternatives. Report an upper bound too when a simple certificate fails, to distinguish an unproven policy from an impossible target.

At the fourth stage, use the legally relevant standard of evidence and the substantive test supplied by the applicable instrument. A model-relative lower welfare guarantee is not a substitute for the authority's burden of proof. Economic certificates can improve transparency and proportionality by explaining a remedy's mechanism, feasible substitutes, and expected effects. They cannot establish jurisdiction or eliminate conditions governing the intervention.

Administration should separate secure verification from unrestricted publication. An independent evaluator can audit sensitive records in a controlled environment and release the model restrictions, code, uncertainty ranges, and aggregate conclusions that are safe and informative. Reproducibility may require authorized re-execution rather than public microdata. This procedure is compatible with the idea of usable evidence while respecting legitimate confidentiality and personal-data constraints.

Review dates and stopping rules should be part of the intervention. A pilot can stop because the certified target is met, because costs exceed the break-even threshold, or because transportability has failed. A later model update should revise both the common rule and its certificate through the registered procedure. Continuing an expensive experiment merely because residual uncertainty remains confuses universal identification with the decision objective.

10. European legal fit and a prospective mandate

The economic starting point differs from existing legal tests without displacing them. Article 102 TFEU concerns abuse of a dominant position within the internal market or a substantial part of it where trade between Member States may be affected (European Union, 2016). The Commission's final exclusionary-abuse Guidelines were adopted on 3 September 2026. Relevant-market definition is generally a prerequisite, subject to their qualifications; dominance, conduct, and justification remain distinct inquiries. Acquiring dominance on the merits is not prohibited (European Commission, 2026). A service boundary or failed comparison certificate cannot bypass those conditions.

The 2024 Market Definition Notice supplies the substitution framework and treats market shares as informative but insufficient on their own (European Commission, 2024). Thus the proposed service boundary organizes evidence for an economic decision while legal market definition identifies the competitive constraints relevant to the applicable law. Their boundaries may overlap or differ. Arguments about effects, efficiencies, or innovation must also respect the instrument's affected-market and beneficiary conditions; an unrestricted social welfare sum is not the governing legal standard.

The Digital Markets Act applies a separate designation regime to specified core platform services of gatekeepers. Article 3 requires significant internal-market impact, an important gateway between business and end users, and an entrenched durable position or a foreseeable such position, with statutory presumptions. Article 1 preserves antitrust rules. Article 6 provides scoped portability, business-user data access, and interoperability obligations. These can affect the modeled experiments, but designation and the particular obligation's scope govern their application (European Union, 2022a).

The Digital Services Act's vetted-research access under Article 40 is directed to understanding Article 34 systemic risks and assessing Article 35 mitigation for the covered services. Delegated Regulation 2025/2050 supplies technical conditions and procedures. This is not a general entitlement to confidential platform records for any competition study (European Union, 2022b; European Commission, 2025). The European Centre for Algorithmic Transparency supplies scientific and technical support for DSA implementation; it is an example of evaluation capacity, not a separate power to mandate every proposed experiment (European Commission, n.d.). These instruments permit different institutional uses. A procurement authority might run a lawful evaluation within its purchasing powers. A competition authority might use the certificate as supplemental economic evidence within a properly established case. A digital regulator might evaluate implementation of a duty that already applies. Each route requires its own authority, facts, and procedural safeguards. This article offers no legal finding concerning an actual firm and no inference of infringement from a statistical impossibility.

A prospective informative-access mandate would therefore need legislation defining its own trigger. One defensible design would authorize a time-limited evaluation order for a specified service only after an authority identifies a consequential public decision, a material evidence deficit, and a credible feasible intervention. The trigger would concern control of the relevant evidence-generating process and potential uninternalized decision benefits. It would not deem every large firm dominant or amend Article 102 through administrative reinterpretation.

The initial order should require a comparison of less burdensome means, including existing records and voluntary or public evaluation. Before imposing a substantive access change, the authority should publish a decision-specific objective, evidence restrictions, expected cost range, and candidate common rules. Where disclosure alone cannot generate a missing contrast, a

tightly scoped prospective trial could be authorized. Safety, privacy, confidentiality, eligibility, and preservation of trusted records would constrain its design.

A legislative scheme would also specify who bears costs, who can challenge model restrictions, and how independent review operates. Public funding could be justified for broadly shared evaluation benefits, but its source and opportunity cost remain in the appraisal. An authorization should expire or narrow when the comparison is no longer useful, the target is met, or the certified return fails. The proposal is an evaluation procedure with conditional remedial powers, rather than a permanent mandate to maintain every competitor or every possible counterfactual. The case for such legislation must be assessed against existing institutional capacity, administrative costs, and private experimentation. The present results support a design question: whether a limited authority to generate interpretable evidence improves consequential decisions where existing law and contracting leave a demonstrable gap. They do not establish the size of that gap. A pilot legislative program with registered evaluation and review would generate evidence about the procedure itself.

11. Limitations and conclusion

The model deliberately isolates a review decision and a finite remedy menu. A complete equilibrium account would model entry, effort, reporting, innovation, and repeated authority action. Those responses can be represented by competing reduced-form models here, but a certificate is only as credible as their counterfactual restrictions. Strategic actors may change behavior once a pilot or audit is announced. Relevant external information may also arrive elsewhere, reducing the marginal value of the intervention.

Normative choices remain visible. The loss function, distributional weights, discounting, target, and ambiguity attitude are not learned from allocation probabilities. A robust lower guarantee can

be conservative and can fail even when some expected-welfare appraisal favors action. Mandatory legal obligations may apply without a positive certificate under this research criterion. Conversely, an attractive certificate does not supply legal authority. These are limits of the economic decision problem rather than reasons to suppress its measurable implications. Informative access can nevertheless be evaluated more sharply than by counting firms or demanding additional data. The authority can identify the decision, specify an interpretable experiment, retain a defensible set of complete models, and ask whether one implementable rule clears a costed threshold. Calibration and comparison may need to be supplied together. The best remedy can be a small audit, a pilot, a narrow interface, or no additional experiment. The framework makes those conclusions reviewable through the assumptions and evidence on which they depend.

Appendix A Proofs and certificate extension

Proof of Proposition 1. The decision space is the product of finitely many probability simplices, one for each signal. It is compact and convex. Each $\Delta_m(r,d)=b_{r,m}-\delta^L \sum_{z,a} h_{r,m}(z,a)d(a|z)$ is affine and continuous in d . Its minimum over finite M is continuous, so the maximum is attained. The constraints on v say exactly that v is no larger than every Δ_m . Maximizing v therefore gives q_r . Finite zero-sum minimax, equivalently linear-program duality, permits interchanging maximization over d and minimization over model weights w . For each signal the weighted-loss minimizing action attains the inner optimum, yielding the stated dual formula.

The dual also clarifies model enlargement. If M is replaced by a superset, the minimum for every rule weakly falls, so q_r cannot increase. If a model is removed only because it makes a remedy unattractive, the certificate changes its evidentiary claim rather than becoming stronger evidence. A documented calibration result can justify removal; the remedy's preferred answer cannot.

Proof of Proposition 2. Write $u=d(1|1)$ and $v=d(1|0)$. Under model $+$, risk is $[1-(2q-1)(u-v)]/2$; under model $-$ it is $[1+(2q-1)(u-v)]/2$. Their sum is one. The maximum is at least one-half, attained by $u=v$. Observing trusted orientation metadata permits $u-v=1$ under $+$ and $u-v=-1$ under $-$ within one rule conditioned on that metadata. Either metadata alone or the uncalibrated signal alone has value zero relative to uninformative baseline, whereas the bundle reduces risk by $q-1/2$.

Proof of Proposition 3. Baseline loss is $\lambda\pi$. Symmetry of the correctly oriented signal gives follow-signal loss $\lambda(1-q)$. Subtracting yields $\Delta S - \Delta C + \delta^L \lambda(\pi+q-1)$. Applying the four lower or upper restrictions gives the displayed bound. A jointly feasible corner attains it. Without such joint feasibility the bound remains sufficient but can be conservative; independent extrema must not be described as the sharp identified value.

Proof of Proposition 4. Let m^* be the true model and let $E=\{m^* \in M(D)\}$. On E , the selected rule's certificate includes m^* , so $\Delta_{m^*}(r(D),d(D)) \geq \tau$. Conditional independence of calibration and the future task given m^* makes this expression the relevant conditional expected gain, including the data-selected rule. Therefore the probability, over calibration evidence, that conditional expected gain is below τ is at most $\Pr(E \text{ complement}) \leq \alpha$. The argument does not turn sampling coverage into coverage of omitted structural models.

For a continuous model class, a finite grid certificate can be extended with a uniform error allowance. Suppose every retained model m has a grid representative g such that $|\Delta_m(r,d) - \Delta_g(r,d)| \leq \varepsilon_r$ for every implementable d . Then a grid certificate at $\tau + \varepsilon_r$ certifies τ for the full class. Subtract ε_r from the grid guarantee and apply the inequality for its representative. Bounds on direct effects and total variation of the joint state-signal laws can supply the allowance for bounded losses; they must include baseline as well as remedy error.

Appendix B Pilot calculation and finite experimentation

For odd n , the correctly oriented majority signal in Section 8 has accuracy $q_n(p)$. Coupling Bernoulli draws by common uniform random variables shows that this majority accuracy increases with p , so p_L supplies its lower bound. Its exact expression is:

$$q_n(p) = \sum_{j=(n+1)/2}^n \frac{n!}{j!(n-j)!} p^j (1-p)^{n-j}.$$

At $p=0.6$, the values for $n=5,9,15,31$ are respectively 0.6825600, 0.7334323, 0.7868968, and 0.8716183. Multiplying the last two values by their stated welfare stakes gives Table 1's unrounded pilot and replication certificates, €0.1032071 million and €0.0694564 million. Under 15 percent arbitrary contamination the fifteen-trial lower accuracy is 0.6688623 and its certificate is –€0.0030239 million. These are deterministic evaluations of the stipulated experiment.

Proof of Proposition 5. Couple n and $n+2$ trials by adding two independent Bernoulli draws to the first n . Majority classification can change only at the two counts nearest the threshold.

Evaluating those events gives, with $n=2k+1$:

$$q_{2k+3}(p) - q_{2k+1}(p) = \frac{(2k+1)!}{k!(k+1)!} [p(1-p)]^{k+1} (2p-1).$$

The ratio of successive positive increments is $2(2k+3)p(1-p)/(k+2)$, strictly below one when $1/2 < p < 1$. Thus I_n 's two-trial increments strictly decrease. Add the next pair exactly while $\delta^L \lambda[q_{n+2}(p_L) - q_n(p_L)]$ exceeds $2(\eta + \kappa)$, allowing either adjacent count at equality. Compare the best positive pilot with baseline to determine participation. Increasing λ or discount weight raises every incremental return, while increasing marginal cost lowers it, giving the stated weak comparative statics.

Proof of Proposition 6. With loss between zero and λ , no rule can reduce undiscounted decision loss by more than λ . If the operating-surplus increment is at most H and implementation-cost

increment is at least κn , every pilot has gain at most $H + \delta^L \lambda - \kappa n$. Once $n > (H + \delta^L \lambda) / \kappa$, this upper bound is negative and the costless baseline dominates. Setup and additional adverse effects tighten the bound. The result is a finite decision-horizon statement, not a claim that universal structural identification has been achieved.

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Declarations

Data availability. This article contains theoretical analysis and transparent hypothetical calculations, with no empirical company or individual-level dataset. Appendix B supplies the exact formula and parameter values needed to reproduce the numerical illustration.

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